

Quick Activity 1: School Supplies Calculator

A notebook costs ₱45, and a pen costs ₱12. A student buys several notebooks and pens and pays a certain amount. Develop an algorithm that calculates the **total cost** and **change**.

Assume that quantities are positive whole numbers and the payment is enough to cover the purchase.

Input	Process	Output
Quantity of notebook, quantity of pens, price of notebook, price of pens	Total notebooks = quantity × price Total pens = quantity × price; total cost = total notebooks + total pens Change = payment – total cost	Total cost, change.

Decomposition:

- Get the price of pens
- Get the price of notebooks
- Get the total cost
- Get the change
- Calculate the quantity and price of the pens
- Calculate the quantity and price of the notebooks
- Add up the price of pens and notebooks
- Get the total payment minus the total cost

PSEUDOCODE:

START

READ quantity, payment

 $\text{pensSubtotal} \leftarrow \text{quantity} \times 12$ $\text{notebookSubtotal} \leftarrow \text{quantity} \times 45$ $\text{totalCost} \leftarrow \text{pensSubtotal} + \text{notebookSubtotal}$ $\text{change} \leftarrow \text{payment} - \text{totalCost}$

DISPLAY totalCost, change

END

DRY-RUN CALCULATIONS:

Step	Variable	Calculation	Result
1	pensSubtotal	2×12	24
2	notebookSubtotal	3×45	135
3	totalCost	$24 + 135$	159
4	change	$200 - 159$	41

Step	Variable	Calculation	Result
1	pensSubtotal	3×12	36
2	notebookSubtotal	2×45	90
3	totalCost	$36 + 90$	96
4	change	$150 - 96$	54